STACKS, DISCRIMINANTS: LEMMA-CHARACTERIZE-INVERTIBLE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2091-2104 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 2106-2126, the complete author proof (19 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 26 of queue_discriminant.json (pb=1012, stmt 2091-2104, proof 2106-2126).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0BVF: sources/stacks-discriminant/pinned-original-source.tex, source lines 590–593. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BVE: sources/stacks-discriminant/pinned-original-source.tex, source lines 555–558. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 080Q: sources/stacks-discriminant/pinned-original-source.tex, source lines 2126–2126. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BVG: sources/stacks-discriminant/pinned-original-source.tex, source lines 649–660. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Invertibility of the relative dualizing module.*

(1) *If $A \to B$ is a quasi-finite flat homomorphism of Noetherian rings, then $\omega_{B/A}$ is an invertible B-module if and only if $\omega_{B \otimes_A \kappa(\mathfrak{p})/\kappa(\mathfrak{p})}$ is an invertible $B \otimes_A \kappa(\mathfrak{p})$-module for all primes $\mathfrak{p} \subset A$.*

(2) *If $Y \to X$ is a quasi-finite flat morphism of Noetherian schemes, then $\omega_{Y/X}$ is invertible if and only if $\omega_{Y_x/x}$ is invertible for all $x \in X$.*

Date: stacks-project:discriminant.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Proof of (1). As $A \to B$ is flat, the module $\omega_{B/A}$ is A-flat, see Lemma 0BVE. Thus $\omega_{B/A}$ is an invertible B-module if and only if $\omega_{B/A} \otimes_A \kappa(\mathfrak{p})$ is an invertible $B \otimes_A \kappa(\mathfrak{p})$-module for every prime $\mathfrak{p} \subset A$, see More on Morphisms, Lemma 080Q. Still using that $A \to B$ is flat, we have that formation of $\omega_{B/A}$ commutes with base change, see Lemma 0BVF. Thus we see that invertibility of the relative dualizing module, in the presence of flatness, is equivalent to invertibility of the relative dualizing module for the maps $\kappa(\mathfrak{p}) \to B \otimes_A \kappa(\mathfrak{p})$.

Part (2) follows from (1) and the fact that affine locally the dualizing modules are given by their algebraic counterparts, see Remark 0BVG. □